Math 131: Complex Analysis

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03/30

1. Let $\mathbb{C} = \mathbb{R}^2$ with vector addition $(a, b) + (c, d) = (a + c, b + d)$ and multiplication $(a, b)(c, d) = (ac - bd, ad + bc)$. Note $\mathbb{R} \hookrightarrow \mathbb{C} : a \to (a, 0)$.

2. $$\begin{aligned} (a, b) &= (a, 0) + (0, 1)(b, 0) \\ \implies (0, 1)^2 &= (-1, 0) \\ \implies (a, b) &= (a, 0) + i(b, 0) \end{aligned}$$

3. Let D be a disk, $F : D \subset \mathbb{R}^2 \to \mathbb{R}^2$ denote a function defined on D taking values in $\mathbb{R}^2$, then $F(x, y) = (u(x, y), v(x, y))$ where $u, v : D \subset \mathbb{R}^2 \to \mathbb{R}$. Then
$$f(x + iy) = u(x, y) + iv(x, y)$$

4. **Proposition 1.7:** Suppose $D \subset \mathbb{R}^2$ is open and connected and $u : D \subset \mathbb{R}^2 \to \mathbb{R}$ such that $u_x \equiv 0 \equiv u_y$, then u is constant.

5. **Definition:** $f : D \subset \mathbb{R}^2 \to \mathbb{R}^2$ is differentiable at an interior point $z_0 \in D$. If there exists a linear map $T : \mathbb{R}^2 \to \mathbb{R}^2$ such that
$$\lim_{z \to 0} \frac{1}{||z||} ||f(z + z_0) - f(z_0) - T(z)|| = 0$$

6. **Remark:** If $f(z) = (u(x, y), v(x, y))$ is differentiable at z_0 then u, v have 1st partials at z_0 and the matrix of T with respect to the standard basis is
$$[T]_\sigma = \begin{pmatrix} u_x(z_0) & u_y(z_0) \\ v_x(z_0) & v_y(z_0) \end{pmatrix}$$
This is the Jacobian. Note that the existence of Jacobian doesn't mean that the fuc is differentiable.

7. **Theorem 1.8:** Suppose that $f : D \to \mathbb{R}^2$ is given by $f(x, y) = (u(x, y), v(x, y))$ and $D_r(z_0) \subset D$. If the first partials of u, v exist in D are continuous at z_0 then f is differentiable at z_0.

8. If u_x is defined on a neighbourhood of z_0 then we can talk about its partials,

$$u_{xy}(z_0) = \frac{\partial}{\partial y}\frac{\partial u}{\partial x}(z_0) = \frac{\partial^2 u}{\partial y \partial x}(z_0)$$

$$u_{xx}(z_0) = \frac{\partial}{\partial x}\frac{\partial u}{\partial x}(z_0) = \frac{\partial^2 u}{\partial x^2}(z_0)$$

9. **Clairaut's(?) Theorem:** If u_{xy} and u_{yx} are continuous at z_0 then $u_{xy}(z_0) = u_{yx}(z_0)$.

10. **Definition:** We define $\exp : \mathbb{C} \to \mathbb{C}$ by

$$\exp(x + iy) = e^x \cos(y) + ie^x \sin(y)$$

we often write e^z instead of $\exp(z)$.

11. **Proposition 1.16:**
    (a) $e^{z+2\pi i} = e^z$
    (b) $e^{z+w} = e^z e^w$
    (c) $e^z = e^w \iff w = z + 2\pi ik,\ k \in \mathbb{Z}$
    (d) $(e^z)^n = e^{nz},\ n \in \mathbb{Z}$

    Proof: Let $z = x + i\theta$ and $w = y + i\phi$ **TODO:** ,

12. $\tau \in \mathbb{C}$ is unimodular if $|\tau| = 1$. This happens $\iff$ $\tau = e^{i\theta}$, $\theta \in \mathbb{R}$. If $z \in \mathbb{C} \setminus \{0\}$ then $z = |z| \cdot \frac{z}{|z|} = re^{i\theta}$ when $r = |z|$ and $\theta \in \mathbb{R}$. We have $\arg(z)\{\theta \in \mathbb{R} : z = |z|e^{i\theta}\}$ and if $z = re^{i\theta_0}$ then,

$$\arg(z) = \{\theta_0 + 2\pi k : k \in \mathbb{Z}\}$$

Note that arg is a set-valued function defined on $\mathbb{C} \setminus \{0\}$.

13. **Proposition 1.22:** Let $f(z) = \exp(z)$. Then f maps $\mathbb{C}$ to $\mathbb{C} \setminus \{0\}$. If $z \neq 0$, then $\{w \in \mathbb{C},\ e^w = z\} = \{\ln(|z|) + i\theta : \theta \in \arg(z)\} = \ln(|z|) + i\arg(z)$. If $z \neq 0$ then $\log(z) = \ln(|z|) + i\arg(z)$.

14. **Definition:** Suppose that f is a set-valued function on a set D. Then a branch of f in D is a continuous function $F : D \to \mathbb{C}$ such that $F(z) \in f(z)$ for all $z \in D$. Example: Let $z \in \mathbb{C} \setminus \{0\}$ then $\exists! \theta \in (-\pi, \pi]$ such that $\theta \in \arg(z)$. Then we can define $\mathrm{Arg} : \mathbb{C} \setminus \{0\} \to (-\pi, \pi]$ such that $\mathrm{Arg}(z) \in (-\pi, \pi]$ and $\mathrm{Arg}(z) \in \arg(z)$ has an discontinuity on $(-\infty, 0]$ but is continuous on $D^\star = \mathbb{C} \setminus (-\infty, 0]$ (See lemma 1.26).

    A branch of arg in D gives us a branch of log in D. We call $\mathrm{Log}(z) = \ln(|z|)_i \mathrm{Arg}(z)$ the principal branch of log in $D^\star$.

15. Branch-cut

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1. If $(a_n) \in \mathbb{C}$ is a sequence then $\sum_{n=1}^{\infty} a_n$ converges if the sequence (S_n) converges, where $S_n = \sum_{i=1}^{n} a_n$, and then,
$$\sum_{n=1}^{\infty} = \lim_{n\to\infty} S_n$$
2. The sequence converges absolutely if $\sum_{n=1}^{\infty} |a_n| < \infty$
3. **Lemma 1.29:** An absolutely convergent series is convergent.
Proof: Suppose that $\sum_{n=1}^{\infty} |a_n| < \infty$. Then given $\epsilon > 0$, $\exists N$ such that for all $n > N$,
$$\sum_{n=N}^{\infty} |a_n| < \epsilon$$
Now if $m > n > N$, we have
$$|S_m - S_n| \leq \sum_{k=n+1}^{m} |a_k| \leq \sum_{k=n}^{N} |a_k| < \epsilon$$
That is (S_n) is Cauchy and (S_n) converges.
4. **Theorem 1.30:** Let $(a_n) \in \mathbb{C}$
   (a) If $|a_n| < b_n$ and $\sum_{n=1}^{\infty} b_n < \infty$ then (a_n) is absolutely convergent.
   (b) If $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = L$, then $\sum_{n=1}^{\infty} a_n$ is absolutely convergent if $L < 1$, and diverges if $L > 1$.
   (c) Let $L = \limsup |a_n|^{1/n}$. If $L < 1$ then $\sum a_n$ converges absolutely and diverges if $L > 1$.
   Note: if $(a_n) \in \mathbb{R}$, then $\alpha_n = \sup\{a_k : k \geq n\}$ and $\alpha_{n+1} \leq \alpha_n$. That is (α_n) is monotonic decreasing.
$$\limsup_n a_n = \lim_n \sup_{k\geq n} a_k = \inf_n \sup_{k\geq n} a_k$$
5. Recall, if (f_n) is a sequence of $\mathbb{C}$-valued functions on $D \subset \mathbb{C}$, then we say (F_n) converges pointwise on D to $F : D \subset \mathbb{C} \to \mathbb{C}$ if
$$\lim_{n\to\infty} F_n(z) = F(z), \qquad \forall z \in D$$
We say that (F_n) converges uniformly to F on D if $\forall\epsilon > 0$ there exists an N such that $\forall n \geq N$,
$$|F_n(z) - F(z)| < \epsilon \qquad \forall z \in D$$
We use the same terminology for the series $\sum_{n=1}^{\infty} F_n$ by considering the partial sums $S_n(z) = \sum_{i=1}^{n} F_i(z)$.

6. **Theorem 1.31 (Weierstrass M-test):** Suppose $D \in \mathbb{C}$ and (F_n) is a sequence of $\mathbb{C}$-valued functions on D such that $|F_n(z)| \leq M_n$ for all $z \in D$. If $\sum_{n=1}^{\infty} M_n < \infty$ then $\sum_{n=1}^{\infty} F_n$ converges uniformlyon D to some $F : D \subset \mathbb{C} \to \mathbb{C}$.

   Proof: Note that $\forall z \in D$, $\sum_{n=1}^{\infty} F_n(z)$ converges to some $F(z) \in \mathbb{C}$ by comparison test. If $\epsilon > 0$, then $\exists N$ such that $\forall n \geq N$, $\sum_{n=N}^{\infty} M_n < \epsilon$.

   Then if $n \geq N$,

$$|F(z) - S_n(z)| = |F(z) - \sum_{k=1}^{n} F_k(z)|$$
$$= \left|\sum_{k=n+1}^{\infty} F_k(z)\right| \leq \sum_{k=n+1}^{\infty} |F_k(z)| \leq \sum_{k=n+1}^{\infty} M_k < \epsilon$$

7. **Definition:** Let $(a_n)_{n=0}^{\infty} \in \mathbb{C}$ and $z_0 \in \mathbb{C}$. Then $K = \sum_{n=0}^{\infty} a_n(z-z_0)^n$ is called a power seires centered at z_0 with coefficients a_k.

8. **Theorem 1.34:** Consider K . Then $\exists R \in [0, \infty]$ such that

   (a) K converges if $|z - z_0| < R$,
   (b) K converges if $|z - z_0| > R$,
   (c) and if $0 < r < R$, K converges uniformly on $D_r(z_0)$.

   Furthermore
$$\frac{1}{R} = \limsup_n |a_n|^{1/n}$$

   TODO: Subdisks?

   Proof: WLOG let $z_0 = 0$. If K converges only on $z = 0$, then we can take $R = 0$ so that (a) and (b) hold and (c) is vacuous.

   Suppose K converges at z_1 with $|z_1| > 0$ then there exists M such that $|a_n z_1^n| \leq M$, for all n.

   If $|z| < |z_1|$ then $|a_n z^n| = |a_n z_1^n| \left|\frac{z}{z_1}\right|^n \leq M \left|\frac{z}{z_1}\right|^n$ and K converges absolutely at z by the comparison test.

   Let $R = \sup\{|z| : K \text{ converges at } z\}$. If $|z| < R$, then $\exists z$ such that K converges at z, and $|z| < |z_1|$. But then K converges absolutely at z.

   If $|z| > R$ then we know K diverges at z.

   Thus (a) and (b) hold with R.

   If $0 < r < R$ then K converges absolutely at $z = r$, and ww can let $M_n = |a_n|r^n$. Therefore K converges uniformly on $D_r(0)$ and (c) holds.

   Now let $R_0^{-1} = \lim_n \sup |a_n|^{1/n}$. If $0 < |z| < R_0$ then $\lim_n \sup |a_n z_n|^{1/n} = R_0^{-1}|z| < 1$. That is, K converges absolutely at z. If $|z| > R_0$ then $\lim_n \sup |a_n z_n|^{1/n} = R_0^{-1}|z| > 1$. That is, K diverges. Hence, $R = R_0$.

9. Recall that a subset $D \subset \mathbb{C}$ is connected if it does not admit a non trivial partition into relatively open sets.

   Remark: A set $A \in \mathbb{C}$ is connected if given open sets U, $V \in \mathbb{C}$ such that $\{A \cap U, A \cap V\}$ is a partition of A (A is a disjoint union of those two sets) and if $A \cap U \neq \emptyset$ then $A \cap V = \emptyset$.

   Definition: We call a connected open subset $D \subset \mathbb{C}$ a domain.

10. If $E \subset \mathbb{C}$ and $z \in E$ then $E(z) = \bigcup\{A \subset \mathbb{C} : z \in A \subset E \text{ and } A \text{ is connected }\}$. Then $E(z)$ is connected. And if $E(z) \cap E(w) \neq \emptyset$ then $E(z) = E(w)$.

    That is $\{E(z) : z \in \mathbb{C}\}$ is a partition of E into connected subsets called the connected components of E.

    Since disks $D_r(z_0)$ are connected it follows that the connected components of an open set U are open. *Exercise W2.1.2*

11. $E \subset \mathbb{C}$ is path connected if given $z, w \in \mathbb{C}$ there is a continuous function $\gamma : [0, 1] \to E$ such that $\gamma(0) = z$ and $\gamma(1) = w$. Because $[0, 1] \subset \mathbb{R}$ is connected, it follows that path connected sets are connected.

    The converse is not true.

Problem . Let $(a_n) \in \mathbb{R}$. Suppose $\sum a_n$ converges.

1. Does $\sum a_n^2$ converge?
2. Does $\sum a_n^3$ converge?

1. **Definition:** Supppose that $F : D \subset \mathbb{C} \to \mathbb{C}$ and that z_0 is an interior point of D. Then F is complex differentiable at z_0 if

$$F'(z_0) = \lim_{h \to 0} \frac{F(z_0 + h) - F(z_0)}{h}$$

exists, then we call $F'(z_0)$ the derivative of F at z_0. If D is open and $F'(z)$ exists for all $z \in D$ then we say that F is holomorphic on D and write $H(D)$ for the set of all holomorphic functions on D. If $F \in H(\mathbb{C})$ then we say F is entire.

Remark: If $F(z) = z^n$ for $n \in \mathbb{N}$, then $f^{-1}(z) = nz^{n-1}$ with the same proof as Math 3. Similarly, the usual sum, product, quotinent and chain rules with basically the same proofs.

Remark: Thus, polynomials are entire, and rational functions are holomorphic on their natural domains.

Example: Let $f(z) = |z|^2$. Thus, if $z = x + iy$ then $f(z) = x^2 + y^2$. It's companion is $F(x, y) = (x^2 + y^2, 0)$.

Recall that $|z|^2 = z\overline{z}$, so

$$\begin{aligned} f'(z) &= \lim_{h \to 0} \frac{|z + h|^2 - |z|^2}{h} \\ &= \lim_{h \to 0} \frac{(z + h)(\overline{z} + \overline{h}) - |z|\overline{z}}{h} \\ &= \lim_{h \to 0} \frac{z\overline{h} + \overline{z}h + h\overline{h}}{h} \\ &= \overline{z} + z \lim_{h \to 0} \frac{\overline{h}}{h} + 0 \end{aligned}$$

Suppose $f'(z)$ exists and $h = x + iy$.

$$\begin{aligned} f'(z) &= \overline{z} + \lim_{x+iy \to 0} \frac{x - iy}{x + iy} \\ &= \overline{z} + z \lim_{x \to 0} \frac{x}{x} = \overline{z} + z \end{aligned}$$

But also must have

$$f'(z) = \overline{z} + \lim_{y \to 0} \frac{-y}{y} = \overline{z} - z$$

Then, $f'(z)$ exists only if $z = 0$ and $f'(0) = 0$.

2. **Theorem 2.4:** Suppose $r > 0$ and

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n \qquad \forall z \in D_r(z_0)$$

Then $f \in H(D_r(z_0))$ and

$$f'(z) = \sum_{n=1}^{\infty} n a_n (z - z_0)^{n-1} \qquad \forall z \in D_r(z_0)$$

Proof: Since $\lim_{n\to\infty} n^{1/n} = 1$, the radius of convergence of $f(z)$ and $f'(z)$ are equal. Thus $\forall z \in D_r(z_0)$ it suffies to see that $f'(z)$ exists and equals $g(z) = \sum_{n=1}^{\infty} n a_n (z - z_0)^{n-1}$.

Using a change of variables we can assume $z_0 = 0$. Fix $z \in D_r(0)$ and p such that $0 \leq |z| \leq p < r$. If $w \neq z$, then

$$\frac{f(w) - f(z)}{w - z} - g(z) = \sum_{n=1}^{\infty} \left(\frac{a_n w^n - a_n z^n}{w - z} - n a_n z^{n-1} \right)$$

where $A_1 = 0$ and for $n \geq 2$

$$\begin{aligned} A_n &= \frac{w^n - z^n}{w - z} - n z^{n-1} \\ &= \left[\sum_{k=0}^{n-1} w^{n-1-k} z^k \right] - n z^{n-1} \end{aligned}$$

We claim this equals,

$$= (w - z) \sum_{k=1}^{n-1} k z^{k-1} w^{n-k-1}$$

to see this, expand.

Thus if $|w| < p$ and $n \geq 2$.

$$\begin{aligned} A_n &\leq |w - z| \sum_{k=0}^{n-1} k p^{n-2} \\ &= |w - z| p^{n-2} \frac{n(n-1)}{2} \end{aligned}$$

Thus if $|w| < p$,

$$\left|\frac{f(w)-f(z)}{w-z}-g(z)\right| \leq |w-z|\sum_{n=2}^{\infty} n^2|a_n|p^{n-2}$$

But

$$\limsup_n |a_n|^{1/n} = \frac{1}{R}$$

and $p < r \leq R$. That implies $\sum_{n=2}^{\infty} n^2|a_n|p^{n-2}$ converges to some $M > 0$. Therefore, $|w| < p$, then

$$\left|\frac{f(w)-f(z)}{w-z}-g(z)\right| \leq |w-z|M$$

then $f'(z)$ exists and equals $g(z)$.

3. **Corollary 2.7:** If

$$f(z) = \sum_{n=0}^{\infty} a_n(z-z_0)^n$$

for all $z \in D_r(z_0)$. then f has derivatives of all orders in $D_r(z_0)$ and

$$a_n = \frac{f^{(n)}(z)}{n!}$$

In particular a_n is uniquely determined by f at z_0.

Proof: By induction

$$f^{(k)}(z) = \sum_{n=k}^{\infty} n(n-1)\cdots(n-k-1)a_n|z-z_0|^{n-k}$$

for all $z \in D_r(z_0)$.

(HW 2.1.8)

4. **Definition:** If f has derivatives of all orders at z_0 then

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!}(z-z_0)^n$$

is called the Taylor Series for f about z_0.

We know that $\frac{1}{1-z} = 1+z+z^2+\cdots, |z| < 1$ Hence, $\frac{1}{1+z^2} = 1-z^2+z^4-z^6+\cdots, |z| < 1$. Note $f^{(2027)}(0) = 0, f^{(2026)}(0) = -(2026!)$.

5. **Definition:** Suppose that $D \subset \mathbb{C}$ is open. We say $f : D \subset \mathbb{C} \to \mathbb{C}$ is analytic on D if given $z_0 \in D$ there exists $r > 0$ such that $D_r(z_0) \subset D$ then $\exists a_n$ such thtat

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$$

for all $z \in D_r(z_0)$. (W Ex. 2.1.7)

6. Given $f : D \subset \mathbb{C} \to \mathbb{C}$,

$$f(x_i y) = u(x, y) + iv(x, y)$$

where $u, v : D \subset \mathbb{R}^2 \to \mathbb{R}$ and let

$$F(x, y) = (u(x, y), v(x, y)).$$

Example: $f(z) = z^3$

$$f(x + iy) = (x + iy)^3 = x^3 + i3x^2y - 3xy^2 - iy^3$$

that is,

$$u(x, y) = x^3 - 3xy^2, \qquad v(x, y) = 3x^y - y^3$$

7. **Cauchy-Riemann Equations:** We say $f(x + iy) = u(x, y) + iv(x, y)$ satisfies the Cauchy-Riemann equations at z_0 if

$$u_x(z_0) = v_y(z_0) \text{ and } u_y(z_0) = -v_x(z_0)$$

If we use the notation $f_x = u_x + iv_x$ and $f_y = u_y + iv_y$ then we get

$$f_x(z_0) = -if_y(z_0)$$

Also,

$$f_x(z) = \lim_{x \to 0,\, x \in \mathbb{R}} \frac{f(z + x) - f(z)}{x} = u_x(z_0) + iv_x(z_0)$$

$$f_y(z) = \lim_{y \to 0,\, y \in \mathbb{R}} \frac{f(z + iy) - f(z)}{y} = u_y(z_0) + iv_y(z_0)$$

8. **Theorem:** Let f and F be as above. Assume z_0 is an interior point of D. Then f is complex differentiable if and only if F is complex differentiable at z_0 and the Cauchy-Riemann equations hold. In the case f' and z_0 exist, then

$$f'(z_0) = f_x(z_0) = -if_y(z_0)$$

06/04

1. **Theorem:** Suppose that $f(x+iy) = u(x,y) + iv(x,y)$ is defined on $D \subset \mathbb{C}$ and let
$$F(x,y) = (u(x,y), v(x,y))$$
be the companion function. If z_0 is an interior point of D then f is complex differentiable at z_0 if and only if the Cauchy-Riemann equations hold and F is differentiable at z_0, if $f'(z_0)$ exists, then
$$f'(z_0) = f_x(z_0) = -ify(z_0)$$
Proof: Suppose $f'(z_0)$ exists, then
$$\begin{aligned} f'(z_0) &= \lim_{x+iy \to 0} \frac{f(z_0 + x + iy) - f(z_0)}{x+iy} \\ &= \lim_{x \to 0} \frac{f(z_0 + x) - f(z_0)}{x} = f_x(z_0) \end{aligned}$$
But also, we have,
$$f'(z_0) = \lim_{y \to 0} \frac{f(z_0 + iy) - f(z_0)}{iy} = -if_y(z_0)$$
The rest is Exercise 2.2.5.

Example: Let $f(z) = \bar{z}$. Then $u(x,y) = x$, $v(x,y) = -y$. Then,
$$u_x \equiv 1, \qquad v_y \equiv 1$$
Cauchy Riemann never holds. That is, the function is not complex differentiable anywhere. But $F(x,y) = (x, -y)$.

2. **Theorem 2.15:** Suppose D is open, and $F : D \subset \mathbb{C} \to \mathbb{C}$ has continuous first partials in D. If the Cauchy-Riemann Equations hold throughout D, then $f \in H(D)$.

Corollary: The complex exponential function $f(z) = \exp(Z)$ is entire and $f'(z) = f(z)$, $\forall z \in \mathbb{C}$.

Proof: $f(x+iy) = e^x \cos(y) + ie^x \sin(y)$. Then, $u(x,y) = e^x \cos(y)$ and $v(x,y) = e^x \sin(y)$. Then,
$$u_x = e^x \cos(y) = v_y, \quad u_y = -e^x \sin(y) = -v_x$$
Hence, f is entire.
$$f'(x+iy) = f_x(x+iy) = e^x \cos(y) + ie^x \sin(y) = f(x+iy).$$

3. **Remark:** If $f(z) = \exp(z)$ then f has derivatives of all orders, and its Taylor series at $z_0 = 0$ is
$$\sum_{n=0}^{\infty} \frac{z^n}{n!}$$

4. **Proposition:** Suppose D is a domain and $f \in H(D)$. Then if $f'(z) = 0$ for all $z \in D$, then f is constant.

   Proof: Let $f(z) = u(z) + iv(z)$. Since
$$f'(z) = f_x(z) = -if_y(z)$$
   it follows that $u_x \equiv u_y \equiv v_x \equiv v_y \equiv 0$. That implies that u, v are constant, which implies f is constant.

5. **Lemma 2.19:** Suppose f is a branch of $\log(y)$ on a domain D. Then $\Re(f(z)) = \ln(|z|)$, $f = H(D)$ and $f'(z) = 1/z, \forall z \in D$.

   Proof: By definition, f is continuous and $\exp(f(z)) = z$, $\forall z \in D$. Then,
$$|z| = |\exp(f(z))| = \exp(\Re(f(z)))$$
   And
$$\ln(|z|) = \Re(f(z))$$
   Now, let $z_0 \in D$. We want to show that
$$\lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0} = \frac{1}{z_0}$$
   Let $w_0 = f(z_0)$ and fix $\epsilon > 0$. Since,
$$\lim_{w \to w_0} \frac{e^w - e^{w_0}}{w - w_0} = e^{w_0} = z_0$$
   Since $e^{w_0} \neq 0$, $\exists \partial' > 0$ such that$0 < |w - w_0| < \partial'$ that implies $e^w - e^{w_0} \neq 0$ and
$$\left| \frac{w - w_0}{e^w - e^{w_0}} - \frac{1}{z_0} \right| < \epsilon$$
   Since f is continuous, $\exists \partial > 0$ such that $0 < |z - z_0| < \partial$ we have
$$|f(z) - f(z_0)| < \partial'$$
   Now if we write $w = f(z)$, then
$$\left| \frac{f(z) - f(z_0)}{z - z_0} - \frac{1}{z_0} \right| = \left| \frac{w - w_0}{e^w - e^{w_0}} - \frac{1}{z_0} \right| < \epsilon$$
   since $0 < |w - w_0| < \partial'$, then $f'(z_0) = \frac{1}{z_0}$.
   Similar to Inverse Function Theorem.

6. **Definition:** Suppose that $u : D \subset \mathbb{R}^2 \to \mathbb{R}$ has continuous second partials on D then u is harmonic on D if

$$\begin{aligned}\Delta u(x,y) &= \frac{\partial^2 u}{\partial x^2}(x,y) + \frac{\partial^2 u}{\partial y^2}(x,y) = 0\\ &= u_{xx}(x,y) + u_y y(x,y) = 0\end{aligned}$$

7. **Proposition:** Suppose D is a domain and that $f(x+iy) = u(x,y) + iv(x,y)$ is in $H(D)$. If u and v have continuous second partials then u and v are harmonic in D.

   Proof: We can use Clairaut's Theorem.

$$u_{xx} = \frac{\partial}{\partial x}u_x = \frac{\partial}{\partial y}v_y = \frac{\partial}{\partial y}v_x = -\frac{\partial}{\partial y}u_y = -u_{yy}$$

   Note $v = \Re(-if)$.

8. **Definition:** If u is harmonic in D then we call v a harmonic conjugate for u in D if

$$f(z) = u(z) + iv(z)$$

   is holomorphic in D.

   Weird Stuff: We don't know if the harmonic conjugate is harmonic, and we don't know if v is the harmonic conjugate for u then u is the harmonic conjugate for v,

9. **Lemma:** Suppose that v and w are harmonic conjugates for u is a domain D. Then $\exists r \in \mathbb{R}$ such that

$$w(z) = v(z) + r, \quad \forall z \in D$$

   Proof: By assumption $f = u + iv$ and $g = u + iw$ are in $H(D)$. Then

$$h = -i(f-g) \in H(D)$$

   But h is real valued and constant.

   Exercise 2.2.12: Show that $u(z) = \ln(|z|) = \frac{1}{2}\ln(x^2+y^2)$ is harmonic in $\mathbb{C} \setminus \{0\}$. But u does not have a harmonic conjugate in $\mathbb{C} \setminus \{0\}$.

   Proof: If v were such a function, then v would still be the harmonic conjugate to u in $D^\star = \mathbb{C} \setminus (-\infty, 0]$. That is $v(z) = \mathrm{Arg}(z) + r$, $\forall z \in D^\star$. But $\mathrm{Arg}(z)$ has a jump discontinuity and hence v cannot be extended to a continuous function.

   Corollary: There is no branch of $\log(z)$ in $\mathbb{C} \setminus \{0\}$.

   $\mathrm{Log}(z) = \ln(|z|) + i\mathrm{Arg}(z)$ is holomorphic in $D^\star = \mathbb{C} \setminus (-\infty, 0]$.

$$\mathcal{L}(z) = \ln(|z|) + i\arg_0(z)$$

   is holomorphic in $D_0^\star = \mathbb{C} \setminus [0, \infty)$.

Definition 0.1. *If X is a topological space, then a curve in X is a continuous function $\gamma : [a, b] \subset \mathbb{R} \to X$. We call $[a, b]$ the parameter space and write $\gamma^\star$ for the image $\gamma([a, b]) \subset X$. If $\gamma(a) = \gamma(b)$ we call γ closed, and we also have $\gamma(s) \neq \gamma(t)$ when $a \leq s < t < b$, then we say γ is a simple closed curve.*

*A simple closed curve is called a **Jordan Curve**.*

If $X \subset \mathbb{C}$ we write, $\gamma(t) = x(t) + iy(t)$. We let $\gamma'(t) = \lim_{h\to 0} \frac{1}{h}(\gamma(t + h) - \gamma(t))$ if it exists. This happens iff $x'(t)$ and $y'(t)$ exist, and then

$$\gamma'(t) = x'(t) + iy'(t)$$

Definition 0.2. *γ is smooth if $\gamma'(t)$ exists and is non-zero for all $t \in [a, b]$.*

Examples:

1. Consider $\gamma_1(t) = t + it^2$ if $t \in [-1, 1]$ (smooth) and $\gamma_2(t) = t^3 + it^6$ for $t \in [-1, 1]$ (not smooth $\gamma_2(0, 0) = 0$).
2. $\gamma(t) = t^3 + it^2$, $t \in [-1, 1]$. That is $y = x^{2/3}$ (not smooth).

Theorem 0.3. *Jordan Curve Theorem. If γ is a Jordan Curve, then $\mathbb{C} \setminus \gamma^\star$ consists of two domains each of which has $\gamma^\star$ as its boundary.*

Remark 0.4. *$\gamma^\star$ is compact, it is contained in some $D_R(0)$ and hence exactly one of the components of $\mathbb{C} \setminus \gamma^\star$ of $\mathbb{C}\gamma^\star$ is bounded and we call that the interior of γ.*

Definition 0.5. *A curve $\gamma : [a, b] \to \mathbb{C}$ is called a path or a contour if there is a partition*

$$P = \{a = t_0 < t_1 < \cdots < t_n = b\}$$

such that $\gamma'(t)$ is continuous on $[t_{k-1}, t_k]$ for $1 \leq k \leq n$. If $D \subset \mathbb{C}$ then we say γ is a path in D if $\gamma^\star \subset D$.

Definition 0.6. *Let $\gamma : [a, b] \to \mathbb{C}$ be a curve. Say $\gamma(t) = x(t) + iy(t)$.*

$$\int_a^b \gamma(t)dt = \int_a^b x(t)dt + i\int_a^b y(t)dt$$

Then if γ' is continuous then

$$\int_a^b \gamma'(t)dt = x(b) - x(a) + i(y(b) - y(a))$$

Proposition 0.7. *Suppose γ and σ are curves defined on $[a, b]$. Then,*

1. $\int_a^b(\gamma(t) + \sigma(t))dt = \int_a^b \gamma(t)dt + \int_a^\nu \sigma(t)dt$

2. *If* $\alpha \in \mathbb{C}$ *then* $\int_a^b \alpha\gamma(t)dt = \alpha\int_a^b \gamma(t)dt$.

3. $\left|\int_a^b \gamma(t)dt\right| \leq \int_a^b |\gamma(t)|dt$.

Proof. 1. Straightforward.

2. Is routine if $\alpha \in \mathbb{R}$. If $\alpha = a + ib$ we can multiply it out and use the definition.

3. Let τ be a unimodular constant such that

$$\left|\int_a^b \gamma(t)dt\right| = \tau\int_a^b \gamma(t)dt$$

Now,

$$\begin{aligned}\left|\int_a^b \gamma(t)dt\right| &= \int_a^b \tau\gamma(t)dt \\ &= \mathrm{Re}\left(\int_a^b \tau\gamma(t)dt\right) \\ &= \int_a^b \mathrm{Re}\left(\tau\gamma(t)\right) dt \\ &\leq \int_a^b |\tau\gamma(t)|dt \\ &= \int_a^b |\gamma(t)|dt\end{aligned}$$

□

Definition 0.8. *If* $\gamma : [a, b] \to \mathbb{C}$ *is a path, then the opposite path* $-\gamma$ *is* $-\gamma : [a, b] \to \mathbb{C}$ *where,*

$$-\gamma(t) = \gamma(b + a - t)$$

Definition 0.9. *If* $z, w \in \mathbb{C}$ *then* $[z, w]$ *is the path with* $\gamma : [0, 1] \to \mathbb{C}$ *such that*

$$\gamma(t) = z + t(w - z)$$

Remark 0.10. $-[z, w] = [w, z]$

Remark 0.11. *If* $\gamma : [a, b] \to \mathbb{C}$ *is a path and* $\gamma(t) = x(t) + iy(t)$, *then there exists a partition* $P = \{a = t_0 < t_1 < \cdots < t_n = b\}$ *such that* $\gamma^\star$ *is continuous on* $[t_{k-1}, t_k]$. *Then* x' *and* y' *are Riemann integrals on* $[a, b]$. *Now if* $\sigma : [a, b] \to \mathbb{C}$ *is any curve, then we can define*

$$\int_a^b \sigma(t)\gamma'(t)dt = \sum_{k=1}^{n}\int_{t_{k-1}}^{t_k} \sigma(t)\gamma'(t)dt$$

Similarly we'll write,

$$\int_a^b |\gamma'(t)|dt = \sum_{k=1}^{n} \int_{t_{k-1}}^{t_k} |\gamma'(t)|dt$$

Definition 0.12. *If $\gamma : [a, b] \to \mathbb{C}$ is a path and if $f \in C(\gamma^\star)$ then the path integral of f along γ is*

$$\int_\gamma f(z)dz = \int_a^b f(\gamma(t))\gamma'(t)dt$$

(Recall a linear combination of line integrals Ex 3.2.6).

If $f, g \in C(\gamma^\star)$ then,

$$\int_\gamma (f(z) + g(z))dz = \int_\gamma f(z)dz + \int_\gamma g(z)dz$$

$$\int_\gamma \alpha f(z)dz = \alpha \int_\gamma f(z)dz$$

$$\int_{-\gamma} f(z)dz = -\int_\gamma f(z)dz$$

Recall $[z, w]$ is given by $\gamma(t) = z + t(w - z)$, $t \in [0, 1]$

$$\int_{[z,w]} f(z)dz = \int_0^1 f(\gamma(t))\gamma'(t)dt = (w - z) \int_0^1 f(z + t(w - z))dt$$

Remark 0.13. *We call $\gamma : [0, 2\pi] \to \mathbb{C}$ given by*

$$\gamma(t) = a + re^{it}$$

the positively oriented circle of radius r centered at a.

Theorem 0.14. *Homer's Cauchy Theorem. Let $\gamma(t) = a + re^{it}$ with $r > 0$ and $t \in [0, 2\pi]$. Then if $n \in \mathbb{Z}$, we have that $\int_\gamma (z - a)^n dz = \begin{cases} 0, & n \neq -1 \\ 2\pi i, & n = -1 \end{cases}$*

If $\sigma(t) = e^\alpha$ with $\alpha \in \mathbb{C}$ then $\sigma'(t) = \alpha e^{\alpha t}$. If $\neq -1$, then,

$$\begin{aligned}\int_\gamma (z - a)^n dz &= \int_0^{2\pi} (re^{it})^n rie^{it} dt \\ &= r^{n+1} i \int_0^{2\pi} e^{i(n+1)t} dt \\ &= ir^{n+1} \frac{e^{i(n+1)t}}{i(n+1)}\Big|_0^{2\pi} = 0\end{aligned}$$

Definition 0.15. *Two paths γ_1 and γ_2 with $\gamma_1^\star = \gamma_2^\star$ are called equivalent if*

$$\int_{\gamma_1} f(z)dz = \int_{\gamma_2} f(z)dz$$

for all $f \in C(\gamma^\star)$.

Lemma 0.16 (3.20)**.** *Suppose $\gamma : [a, b] \to \mathbb{C}$ is a path and that $\phi : [c, d] \to [a, b]$ is differentiable with $\phi(c) = a$ and $\phi(d) = b$. Let $\sigma(t) = \gamma(\phi(t))$. Then σ is path defined on $[c, d]$ which is equivalent to γ.*

Proof. Check $\sigma'(t) = \gamma'(\phi(t))\phi'(t)$.
If $f \in C(\gamma_1^\star)$, then

$$\begin{aligned}\int_\sigma f(z)dz &= \int_c^d f(\sigma(t))\sigma'(t)dt \\ &= \int_c^d f(\gamma(\phi(t)))\gamma'(\phi(t))\phi(t)dt\end{aligned}$$

Let $u = \phi(t)$, $du = \phi'(t)$

$$\begin{aligned}\int_\sigma f(z)dz &= \int_{\phi(c)}^{\phi(d)} f(\gamma(u))\gamma'(u)du \\ &= \int_a^b f(\gamma(u))\gamma'(u)du \\ &= \int_\gamma f(z)dz\end{aligned}$$

□

Corollary 0.16.1. *Every path is equivalent to the path defined on $[0, 1]$.*

Definition 0.17. *Suppose $\gamma : [a.b] \to \mathbb{C}$ and $\sigma : [c, d] \to \mathbb{C}$ are paths so that $\gamma(b) = \sigma(c)$. Then their join is the path*

$$\gamma + \sigma : [0, 1] \to \mathbb{C}$$

given by

$$\gamma + \sigma(t) = \begin{cases} \tilde{\gamma}(t), & 0 \le t \le \frac{1}{2} \\ \tilde{\sigma}(t), & \frac{1}{2} \le t \le 1 \end{cases}$$

where $\tilde{\gamma}$ is a path on $\left[0, \frac{1}{2}\right]$ equivalent to γ and $\tilde{\sigma}$ is any path on $\left[\frac{1}{2}, 1\right]$ equivalent to σ.
We could be definite and write

$$\gamma + \sigma(t) = \begin{cases} \gamma(a + 2(b - a)t), & 0 \le t \le \frac{1}{2} \\ \gamma(c + 2(d - c)(t - \frac{1}{2})), & \frac{1}{2} \le t \le 1 \end{cases}$$

And

$$\begin{aligned}\int_{\gamma+\sigma} f(z)dz &= \int_{\tilde{\gamma}} f(z)dz + \int_{\tilde{\sigma}} f(z)fz \\ &= \int_{\gamma} f(z)dz + \int_{\sigma} f(z)dz\end{aligned}$$

Lemma 0.18 (3.7)**.** *Suppose* $\gamma : [a, b] \to D$ *is a path in a domain* D *and* $f \in H(D)$*. Suppose that* f' *is continuous and* γ' *is continuous then* $\sigma(t) = f(\gamma(t))$ *is a path on* $[a, b]$ *such that*

$$\sigma'(t) = f'(\gamma(t))\gamma'(t)$$

Proof. Because f' is continuous, we just need to show that the following equation holds,

$$\sigma'(t) = f'(\gamma(t))\gamma'(t).$$

Let $t \in (a, b)$. Then,

$$\begin{aligned}\sigma'(t) &= \lim_{h\to 0} \frac{1}{h} \left(\sigma(t + h) - \sigma(t)\right) \\ &= \lim_{h\to 0} \frac{1}{h} \left(f(\gamma(t + h)) - f(\gamma(t))\right) \\ &= \lim_{h\to 0} G(h) \frac{\gamma(t + h) - \gamma(t)}{h}\end{aligned}$$

when,

$$G(h) = \begin{cases} \frac{f(\gamma(t+h))-f(\gamma(t))}{\gamma(t+h)-\gamma(t)}, & \gamma(t + h) \neq \gamma(t) \\ f'(\gamma(t)), & \gamma(t + h) = \gamma(t)\end{cases}$$

It is not hard to see that G is continuous ar 0. Now we are done. ☐

Theorem 0.19 (Fundamental Theorem for Path Integrals)**.** *Suppose* D *is a domain and that* $f : D \subset \mathbb{C} \to \mathbb{C}$ *is continuous. Suppose that* $F \in H(D)$ *and* $F'(z) = f(z), \forall z \in D$*. Then if* $\gamma : [a, b] \to D$ *is any path in* D*,*

$$\int_{\gamma} f(z)dz = F(\gamma(b)) - F(\gamma(a))$$

Proof. We have

$$\begin{aligned}\text{LHS} &= \int_a^b f(\gamma(t))\gamma(t)dt \\ &= \sum_{k=1}^{n} \int_{t_{k-1}}^{t_k} f(\gamma(t))\gamma'(t)dt\end{aligned}$$

where γ' is continuous on $[t_{k-1}, t_k]$. But,

$$\frac{d}{dt}(t \to F(\gamma(t))) = f(\gamma(t))\gamma'(t) = \sum_{k=1}^{n} F(\gamma(t_k)) - F(\gamma(t_{k-1}))$$

□

Definition 0.20. *Let* $\gamma : [a, b] \to \mathbb{C}$ *is a path, then the length of* γ *is*

$$\ell(\gamma) = \int_a^b |\gamma'(t)| dt$$

See Remark 3.2.7.

Proposition 0.21. *Suppose* $\gamma : [a, b] \to \mathbb{C}$ *is a path and* $F \in C(\gamma^\star)$.

(a) *If* $|f(z)| \leq M$ *for all* $z \in \gamma^\star$, *then*

$$\left|\int_\gamma f(z)dz\right| \leq M\ell(\gamma)$$

(b) *If* $(f_n) \subset C(\gamma^\star)$ *and if* $f_n \to f$ *uniformly on* $\gamma^\star$, *then*

$$\lim_{n\to\infty} \int_\gamma f_n(z)dz = \int_\gamma f(z)dz$$

Proof. (a)

$$\begin{aligned}\left|\int_\gamma f(z)dz\right| &= \left|\int_a^b f(\gamma(t))\gamma'(t)dt\right| \\ &\leq \int_a^b |f(\gamma(t))|\,|\gamma(t)|dt \\ &\leq M\int_a^b |\gamma'(t)|dt \\ &= M\ell(\gamma)\end{aligned}$$

(b) Let $\epsilon > 0$. Then $\exists N$ such that for all $n \geq N$

$$|f_n(z) - f(z)| < \frac{\epsilon}{\ell(\gamma)+1}, \quad \forall z \in \gamma^\star$$

Now using part (a)

$$\left|\int_\gamma f(z)dz - \int_\gamma f_n(z)dt\right| = \left|\int_\gamma (f(z)dz - f_n(z))dz\right| \leq \frac{\epsilon}{\ell(\gamma)+1}\ell(\gamma) < \epsilon$$

□

Theorem 0.22 (3.31)**.** *Supppose that* γ *is a path and that* $f \in C(\gamma^\star)$*. Let* $D = \mathbb{C} \setminus \gamma^\star$*. Define,*

$$F(z) = \int_\gamma \frac{f(w)}{w-z} dz, \qquad \forall z \in D$$

Then F *is analytic on* D*. For* $n \geq 0$*,*

$$F^{(n)}(z) = n! \int_\gamma \frac{f(w)}{(w-z)^{n+1}} dw$$

Moreover, the radius of convergence of the Taylor series in F *about* $a \in D$ *is at least* $\sup\{R : D_R(a \subset D)\}$*.*

Proof. Fix $a \in D$. Since D is open, $\exists R > 0$ such that $D_R(a) \subset D$. If $z \in D_R(a)$ and $w \in \gamma^\star$ then,

$$\left| \frac{z-a}{w-a} \right| \leq \frac{|z-a|}{R} < 1$$

Hence,

$$\begin{aligned} \frac{1}{w-z} &= \frac{1}{w-a-(z-a)} = \frac{1}{w-a} \frac{1}{1-\frac{z-a}{w-a}} \\ &= \frac{1}{w-a} \sum_{n=0}^{\infty} \left(\frac{z-a}{w-a} \right)^n \\ &= \sum_{n=0}^{\infty} \left(\frac{(z-a)^n}{(w-a)^{n+1}} \right) \end{aligned}$$

Note that if $w = \gamma^\star$, then $|w-a| \geq R$ and $\exists M$ such that $|f(w)| \leq M$, for all $w \in \gamma^\star$. Hence,

$$\left| f(w) \frac{(z-a)^n}{(w-a)^{n+1}} \right| \leq \frac{M}{R} \left(\frac{|z-a|}{R} \right)^n$$

This for fixed $z \in D_R(a)$,

$$\frac{f(w)}{w-z} = \sum_{n=0}^{\infty} f(w) \frac{(z-a)^n}{(w-a)^{n+1}}$$

uniformly in w by Weierstrass m-test.

Thus $\forall z \in D_R(a)$, we have

$$\begin{aligned} F(z) &= \int_\gamma \frac{f(w)}{w-z} dw \\ &= \int_\gamma \left(\sum_{n=0}^{\infty} f(w) \frac{(z-a)^n}{(w-a)^{n-1}} \right) dw \\ &= \sum_{n=0}^{\infty} \left(\int_\gamma \frac{f(w)}{(w-a)^{n-1}} dw \right) (z-a)^n \end{aligned}$$

Thus F is analytic and

$$\frac{F^{(n)}(a)}{n!} = \int_\gamma \frac{f(w)}{(w-a)^{n+1}} dw$$

$\square$

Definition 0.23. *If γ is a closed path and $z_0 \in \mathbb{C} \setminus \gamma^\star$, then the index of γ about z_0 is*

$$\mathit{Ind}_\gamma(z_0) = \frac{1}{2\pi i} \int_\gamma \frac{1}{z - z_0} dz.$$

Note that the Antiderivate of $\frac{1}{z-z_0}$ is

$$\ln(|z - z_0) + i \arg(z - z_0)$$

$$\mathit{Ind}_\gamma(z_0) = \frac{1}{2\pi} \left(\arg(\gamma(b) - z_0) - \arg(\gamma(a) - z_0) \right)$$